Logan M. Edwards

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EDUCATION

Florida Institute of Technology – Melbourne, FL Aug 2024 – Present
B.S. in Astronomy & Astrophysics | Second Major: Planetary Science | Minor: Sustainability
GPA: 3.6 | Anticipated Graduation: May 2027
Relevant Coursework: Observational Astronomy, Methods & Instrumentation for Astronomy, Deep Learning (CNNs & PyTorch), Classical Mechanics, Partial Differential Equations, Modern Physics, Nanotechnology Lab, Remote Sensing, Introduction to Aerospace Engineering, Economics

University of Oxford (Study Abroad) – Oxford, UK Jun 2025 – Aug 2025
Coursework: Differential Equations, Scientific & Technical Communication

RESEARCH & TECHNICAL PROJECTS

Constraints on the Origin of the Asteroid Belt's Compositional Gradient from Debiased Taxonomic Data

Single-author research paper (2026), Department of Aerospace, Physics, and Space Sciences, Florida Institute of Technology
Python, NumPy, SciPy, pandas, AASTeX

- Tested whether the main belt switches sharply from dry S-types to hydrated C-types at the nebular snow line, using the 161,659 main-belt asteroids with a published taxonomy.
- Collapsed collisional families and used inverse-probability weighting to correct the brightness bias that favors S-types, moving the S/C crossover from 2.84 au to 2.41–2.47 au and showing a 10-to-90% transition wider than the belt itself.
- Showed that a static condensation front cannot produce the corrected gradient and that measured albedos reproduce it independently of taxonomy, so the results favor implantation of carbonaceous bodies from beyond Jupiter.
- Packaged the analysis so every figure and data-derived number in the paper regenerates deterministically from two public inputs, with CI on Linux and Windows checking the paper's tables against the results.

Stellar Spectral Classification from Pan-STARRS Imagery

Independent deep learning project | Python, TensorFlow, Gaia DR3, Pan-STARRS1, Hipparcos, WISE

- Designed a multi-task convolutional neural network that predicts Morgan–Keenan spectral classes (O/B/A/F/G/K/M), class×subclass combinations, and evolutionary stage from Pan-STARRS RGB image cutouts.
- Assembled a 96,039-star dataset by querying Gaia DR3, Hipparcos, Pan-STARRS1, and WISE for coordinates, magnitudes, colors, and spectral labels, applying magnitude-aware stretching and automated quality gates.
- Achieved 84.3% top-1 and 98.4% top-2 class accuracy across the seven-class softmax head, with 67,227/14,406/14,406 train/validation/test splits and class balancing to correct steep O- and M-type underrepresentation.
- Found that most misclassifications fall between adjacent H–R diagram pairs (A/F, F/G, K/M), which indicates the network learned the stellar temperature sequence.
- Wrote a technical report documenting catalog ingestion, quality gating, architecture, and per-class evaluation.

Drone Target Identification and Classification System

Independent deep learning project | Python, TensorFlow, YOLO, OpenCV, Computer Vision

- Developed a detection system that pairs a YOLO segmentation front end with a multi-output CNN classifier for real-time target identification in drone and camera footage.
- Consolidated 182,413 images from multiple public sources into a unified, hash-deduplicated dataset spanning 11 fine-grained classes and 3 coarse validity categories, with stratified 70/15/15 splits and per-label caps.
- Trained a regularized convolutional network (32→64→128 channel progression, dual classification heads) using tf.data pipelines, augmentation, learning-rate scheduling, and mixed-precision training.
- Reached 87.1% fine-class (0.836 macro F1) and 91.3% coarse accuracy, against baselines of 23.5% and 34.5%.
- Wrote a technical report covering dataset construction, deduplication, architecture, and class-level evaluation.

Ortega Telescope Exposure Time Calculator

Observational instrumentation utility | Python, JavaScript, Photometry

- Engineered a photometric exposure-time calculator that derives single-frame and total integration times from target magnitude, desired signal-to-noise ratio, seeing, sky brightness, and airmass via a closed-form quadratic solution.
- Modeled all five Johnson–Cousins filters (U, B, V, R, I; 3656 to 8797 Å) with per-filter extinction coefficients ranging 0.60 to 0.08, plate scale, pixel geometry, and 2.9 e^-/pixel/s read noise.
- Indexed sky background across five lunar-age tables (0 to 14 days) and migrated the tool from a Google Colab script used during observing runs to an interactive browser-based calculator.

Asteroid Mining Profitability Pipeline (economicspace)

Independent research pipeline | Python, Streamlit, Docker

- Built an end-to-end pipeline that estimates the net profit of an uncrewed autonomous mining and sample-return mission for every asteroid with usable data, across a 1.58-million-body catalog.
- Chained four stages: a cross-matched asteroid catalog (JPL SBDB, SsODNet, NEOWISE), mineral pricing from recorded metal quotes, cited launch and propulsion cost tables, and a rocket-equation cost cascade.
- Ranked seven delivery destinations from Earth's surface to Mars and found no mission yet profitable: the best case, delivery to the Martian surface, still costs 1.09 times its revenue.
- Compiled a public evidence registry (General_Research) of more than 1,100 tiered sources across 30 domains that backs the pipeline's prices, costs, and physical constants.
- Delivered a Streamlit dashboard, a command-line interface, and a Docker image, with verification that every host reproduces the published results bit for bit.
- Published the catalog, pricing, and cost stages as standalone Python packages with their own versioned releases (AsteroidCatalog, spacevalue, spacecost; see Open-Source Software).

Mars Global Mosaic: Mapping Lava Flows, Fluvial Channels, and Impact Craters

OCN 4704 Remote Sensing, Florida Institute of Technology (Fall 2026) | ArcGIS Pro, ArcPy, GDAL, Python, U-Net, SVM

- Co-registered Viking color, THEMIS day and night thermal infrared, and the HRSC/MOLA elevation model across three coordinate frames onto one grid covering ±60° latitude (86.6% of the surface).
- Mapped seven landform classes with SVM and U-Net classifiers trained on hand-drawn labels and scored on held-out polygons, with Ius Chasma and Athabasca Valles as detailed type areas.
- Detected 465,354 closed depressions in the elevation model, 63.2% of those 5 km and larger matching cataloged craters, and showed that the THEMIS mosaics track calibrated thermal inertia only weakly.

Satellite Space Flight: 3-D Orbit Simulator

Independent simulation project | Python, NumPy, SGP4, MATLAB, pytest

- Implemented an Earth-centered orbit simulator with J_2–J_4 gravity, atmospheric drag, Sun/Moon third-body forces, and solar radiation pressure, integrated with adaptive Dormand–Prince, RK4, or Wisdom–Holman schemes.
- Imported operational satellites from CelesTrak two-line element sets, initialized through SGP4, and generated Walker constellations.
- Modeled multi-stage rocket launches from any point on Earth, launch windows, and delta-v budgets, and validated the physics with a pytest suite against textbook cases and analytic solutions.
- Ported the propagators and force models to a MATLAB package for scripted scenario analysis and plotting.

OPEN-SOURCE SOFTWARE

AsteroidCatalog: Merged, provenance-tagged catalog of 1,580,325 asteroids cross-matched from four public sources (JPL SBDB, SsODNet, NEOWISE, MP3C), with cited per-taxonomy composition estimates and versioned, checksummed data releases.

spacevalue: Prices every mineral and element an asteroid composition can name at eight delivery destinations, from Earth's surface to Mars, as CSV releases that rebuild byte for byte from recorded metal quotes.

spacecost: Cited reference tables for space-mission cost and mass estimation: launch vehicles, propellants with derived tankage, delta-v segments, operational costs, and storage systems.

samebytes: Reproducibility toolkit that verifies a working tree against the Git index, confirms that a host reproduces recorded floating-point results bit for bit, and compares pandas outputs across runs.

PROFESSIONAL EXPERIENCE

AspireCURES – Melbourne, FL

Chief Technology Officer — Aug 2025 – Present

- Direct all web development and technical operations for a private company that matches patients affected by rare white matter diseases and leukodystrophies with clinical trials, research studies, genetic testing, specialists, and advocacy opportunities.
- Led the migration of the patient-matching platform from Squarespace to Cloudflare Pages, rebuilding it in roughly 15,000 lines of hand-written HTML/CSS/JavaScript with serverless patient registration through Pages Functions, Cloudflare KV as a durable write backup, and Zoho CRM for lead handoff.
- Oversee the platform end to end, from patient-facing pages and registration to analytics, deployment, and content automation.
- Created nine disease-specific program pages covering AARS1/AARS2, CSF1R-ALSP/BANDDOS, Nasu–Hakola, Alzheimer's disease, Lewy body dementia, frontotemporal dementia, multiple sclerosis, Huntington's disease, and Friedreich's ataxia, each tracking current research and recruiting trials.
- Automated a twice-weekly research feed that pulls from Europe PMC, PubMed, and ClinicalTrials.gov and publishes AI-assisted summaries of new papers and trials to each disease page and its own RSS feed.

Sisters' Hope Foundation – Melbourne, FL

Vice President of Database Management — May 2025 – Present

- Manage and maintain the organization's patient database, overseeing structural updates, record management, and data verification for a 501(c)(3) global nonprofit serving families affected by rare neurological disease.
- Migrated the foundation's 167-page website from Squarespace to a static site on Cloudflare Pages, with a redirect map that keeps legacy URLs working, a serverless contact form, consent-gated analytics, and CI-enforced link, markup, SEO, and content-parity checks.

Program Support Associate — Jun 2024 – May 2025

- Maintained internal databases and streamlined data workflows, producing reports for program planning and evaluation.
- Led the integration and modernization of the patient database, earning promotion to Vice President.

Intern — Jun 2021 – Jun 2024

- Supported fundraising and outreach, provided general technical support, and assisted patients and their families.

TECHNICAL SKILLS

Programming Languages: *Advanced:* Python; *Proficient:* JavaScript, HTML/CSS, LaTeX; *Intermediate:* Bash, PowerShell, Perl, MATLAB

Libraries & Frameworks: PyTorch, TensorFlow/Keras, scikit-learn, Ultralytics YOLO, OpenCV, NumPy, SciPy, pandas, PyArrow, Astropy, astroquery, Matplotlib, seaborn, Plotly, Pillow, Streamlit, pytest

Machine Learning: Deep Learning, Convolutional Neural Networks, Multi-Task Learning, U-Net Segmentation, Object Detection, Computer Vision, Support Vector Machines, Data Augmentation, Class Balancing, Learning-Rate Scheduling, Mixed-Precision Training, Dataset Curation & Deduplication, Model Evaluation

Astronomy & Data Analysis: Observational Astronomy, Photometry, Telescope & CCD Instrumentation, Astronomical Data Reduction, Stellar Spectral Classification, Astronomical Catalogs (Gaia DR3, Pan-STARRS1, Hipparcos, WISE, JPL SBDB, SsODNet, NEOWISE), Catalog Cross-Matching, Asteroid Taxonomy, Statistical Data Analysis, Data Modeling

Statistics: Selection-Bias Correction (Inverse-Probability Weighting), Bootstrap Resampling, Uncertainty Estimation, Stratified Sampling, Precision/Recall/F1 Evaluation

Orbital Mechanics & Simulation: Orbit Propagation (SGP4, Two-Line Elements), Numerical Integration (Dormand–Prince, RK4, Wisdom–Holman Symplectic), Perturbation Modeling (J_2–J_4, Drag, Third-Body, Solar Radiation Pressure), Launch Modeling, Delta-v Budgets, Rocket-Equation Cost Modeling

GIS & Remote Sensing: ArcGIS Pro, ArcPy, GDAL, File Geodatabases & Shapefiles, GeoTIFF Rasters, Raster Co-Registration & Reprojection, Planetary Image Mosaics (Viking, THEMIS, HRSC/MOLA), Thermal Infrared Imagery, Terrain Analysis, Crater Detection, Land-Cover Classification

Research Software Practice: Reproducible Pipelines, Bit-for-Bit Result Verification, Automated Testing (pytest), Continuous Integration, Linting (Ruff), Python Packaging (pyproject) & CLIs, Semantic Versioning & Changelogs, Versioned Data Releases, Data Provenance & Checksums, Parquet & CSV Data Formats, Technical Writing (LaTeX/AASTeX)

Laboratory Instrumentation: Transmission Electron Microscopy (TEM), Scanning Electron Microscopy (SEM), Atomic Force Microscopy (AFM), Scanning Tunneling Microscopy (STM)

Systems & Server Administration: Linux (Ubuntu, Arch Linux), Windows, Server Management, Linux System Administration, SSH & Remote Administration, Shell Scripting, Docker, Git & Version Control, DNS & Domain Management, Process Automation

Web & Data Infrastructure: Cloudflare Pages & Functions, Cloudflare KV, GitHub Pages, Node.js, Serverless APIs, RSS Feeds, REST API Integration (Europe PMC, PubMed, ClinicalTrials.gov), Content Security Policy, SEO & Structured Data (JSON-LD), Consent-Gated Analytics, Web Accessibility, Database Administration, Database Design, Zoho CRM & CRM Integration

Scientific Disciplines: Planetary Science, Astrodynamics, Space Mission Economics, Environmental Science, Scientific & Technical Communication